

[Home](#) / [My courses](#) / [Cloud Applications Architecture](#) / [Lab](#) / [Lab Exam 2025-2026](#)

Status	Finished
Started	Thursday, 15 January 2026, 2:50 PM
Completed	Thursday, 15 January 2026, 3:50 PM
Duration	1 hour
Grade	9.75 out of 10.00 (97.5%)

Question **1**

Correct

Mark 0.50 out of 0.50

What is CloudFormation?

Select one:

- ☐ a. PaaS Service
- ☐ b. IaaS Service
- ☐ c. None of the above
- ☒ d. IaC Service ✓

Your answer is correct.

Question **2**

Correct

Mark 0.50 out of 0.50

An EC2 instance can be secured by using  which  .

Your answer is correct.

Question **3**

Partially correct

Mark 0.25 out of 0.50

Why would you specify the maximum capacity for an ASG to be greater than the desired capacity?

Select one or more:

- ☒ a. It gives you more room for a rolling upgrade 
- ☐ b. It allows you to manually add instances created outside of the ASG
- ☐ c. They are always the same
- ☐ d. The ASG takes into consideration only the desired capacity. The min and max capacities are deprecated

Your answer is partially correct.

You have correctly selected 1.

Question **4**

Correct

Mark 0.50 out of 0.50

How does ECS ensure the cluster has enough computing capacity when running on EC2?

Select one:

- ☒ a. It leverages an ASG ✓
- ☐ b. You have to manually scale the cluster
- ☐ c. It has its own instance orchestrator
- ☐ d. ECS only works with Fargate, not with EC2

Your answer is correct.

Question **5**

Correct

Mark 0.50 out of 0.50

Your application tier is running in an Auto Scaling Group and you need to change the instance type. In which of the following area can this be achieved?

Select one:

- ☐ a. Auto scaling policy configuration
- ☐ b. Auto scaling tags configuration
- ☒ c. Auto scaling launch template/configuration ✓
- ☐ d. Auto scaling AMI

Your answer is correct.

Question **6**

Correct

Mark 0.50 out of 0.50

AWS EC2 is a service designed for

Select one:

- ☒ a. Creating virtual machines that can act as servers ✓
- ☐ b. Running applications without having to maintain the OS and packages
- ☐ c. Creating highly available managed databases

Your answer is correct.

Question **7**

Correct

Mark 0.50 out of 0.50

An AWS Region is

Select one:

- ☐ a. An organizational unit designed to serve clients from a specific area
- ☐ b. An AWS data center
- ☒ c. An independent and isolated cloud deployed in a specific geographical region ✓

Your answer is correct.

Question **8**

Correct

Mark 0.50 out of 0.50

Due to regulatory requirements, your team must inspect all packets that flow into your system. What load balancer can help you achieve this?

Select one:

- ☒ a. Gateway Load Balancer ✓
- ☐ b. None. This can only be achieved using VPC features
- ☐ c. Network Load Balancer
- ☐ d. Application Load Balancer

Your answer is correct.

Question **9**

Correct

Mark 0.50 out of 0.50

Your team is sharing an AWS VPC with another team so you configured a dedicated subnet with the CIDR 172.31.1.0/28. Your application is running on EC2 instances managed by an ASG. As the demand for your app increases, you notice that new EC2 instances are not created even though the ASG is correctly configured. What is the most probable cause?

Select one:

- ☐ a. You've set a budget limit that's preventing additional costs
- ☐ b. AWS ran out of capacity for that specific region
- ☐ c. The load balancer has reached the maximum number of targets
- ☒ d. There are no more IP addresses available in the subnet ✓

Your answer is correct.

Question **10**

Correct

Mark 0.50 out of 0.50

Which practice BEST aligns with the **Reliability** pillar of the AWS Well-Architected Framework?

Select one:

- ☒ a. Design applications to automatically recover from failures ✔️ Correct! Automatic recovery, fault tolerance, and graceful handling of failures are core principles of the Reliability pillar.
- ☐ b. Use dashboards to monitor application logs and metrics
- ☐ c. Continuously reduce costs by rightsizing resources
- ☐ d. Encrypt all sensitive data stored in Amazon S3

Your answer is correct.

Question **11**

Correct

Mark 0.50 out of 0.50

How does ECS know how to run a container/task?

Select one:

- ☐ a. Based on the provided Dockerfile
- ☐ b. ECS will run all images from the configured ECR repository
- ☐ c. Based on the images you mention during the cluster creation
- ☒ d. Based on the configured task definition ✔️

Your answer is correct.

Question **12**

Complete

Mark 3.00 out of 3.00

Your organization runs a business-critical human resource management system (HMRS) on its infrastructure on-premises. Since the platform has been growing exponentially, the stakeholders decided to migrate it to AWS. You are the architect in charge. You are responsible for choosing the right services and designing the new system on AWS.

The HRMS enables managers and HR personnel to publish job openings and manage employees. The system stores multiple types of documents, including CVs, contracts, and pictures. Other functional requirements include:

- job applications go through a review process
- applicants are notified via email when there is a change in the application process
- the system exposes an API allowing third-party job boards to fetch your job openings
- the system runs a complex algorithm that leverages ML to filter and match job applicants to the optimal roles.

The HRMS is composed of several backend modules written in Java and Node.js. They store data in a MySQL database. The frontend is built with React.

Your team has been struggling lately to run the system locally due to its distributed nature and increasing dependencies. Use this migration to improve this aspect, too.

This is your organization's core product, so the design's primary focus must be the availability of the system and reducing operational/maintenance overhead. Of course, the cost must be optimized whenever possible.

Describe your solution below. Mention which AWS services you would choose, how they communicate with each other, and, equally important, why you have chosen a particular service. Be as explicit as possible.

Optionally, you can upload images to illustrate your design (feel free to use any drawing or diagram tool you wish; images don't have to be pretty).

System Design:

We need to build HMRS on AWS for managers, HR personnel, employees. (move from on premises software to AWS)

Key requirements:

- A. job application go trough process
- B. notification via mail
- C. third party api fetching jobs opening
- D. ml algorithm to match optimal row,
- E. store documents, including CVs, contracts, and pictures

F. reduce operational/maintenance overhead.

1) USERS & AUTHENTICATION

1.A) Cognito User Pool + JWT (since we have login/users/different accounts: managers, employee, hr)

- Use Amazon Cognito user pool for end-user(managers, employee, hr) sign-up/sign-in and jwt tokens
- Frontend receives JWT and sends it in API calls (Authorization: Bearer <token>).

Why not custom auth in DB?

- Custom auth means we must implement reset/MFA/token issuing correctly which means => high security risk and time

Also Why not IAM?

- IAM is for AWS sdev access, not SaaS customers which we don't need in this HMRS application

2) FRONTEND

2.A) S3 Static Hosting (since we have React frontend)

- host the compiled frontend in S3.

Why not EC2/ECS?

- static files don't need always-on compute; servers would be wasted cost/ops, so we leverage the costs

2.B) CloudFront CDN (as we scale, this app will be run in multiple country, regions)

- just put CloudFront in front of S3 to cache content at edge globally, all over the world

Why not only S3?

- S3 is region based while cloudFront improves global latency and most importantly it can handle spikes

(!!!optional -> but its a great idea for this case!!!) 2.C) Private S3 + CloudFront-only access

- make S3 private; allow access only via CloudFront (origin access control/identity).

Why?

- since it prevents bypassing CDN and reduces direct public exposure and it makes it more safe

3) API ENTRY POINT

3.A) API Gateway

- Use API Gateway as the public API "front door" for endpoints like fetching
- we must validate JWT (Cognito authorizer) before requests reach backend.

Why not only ALB?! (system exposes an API allowing third party-job)

- api Gateway is built for API management: auth integration, usage plans, versioning

4) COMPUTE (BACKEND + FOR WORKERS)

4.A) ECS Fargate + ECR (Node/Java services, reduce ops)

- Containerize JAVA, NODE, etc as Docker images; store in ECR; run on ECS Fargate.

Why this, not EC2? AWS manages servers so lower ops, good for small team.

(BOTH API service + heavy background jobs:)

4.D) Split

- API service on Fargat and Heavy workers on fargate pulling from SQS.

IMPORTANT : FOR THE ML WORKERS WE WILL USE **FARGATE SPOT** SINCE IT SAVES **70%** COST TIME

5) DATA STORES

S3 + MySQL

MYSQL will keep the metadata, while we use and leverage the power of S3 Databases to keep heavy documents.

MYSQL:

example of table to handle

ID | Foreign KEY| name | S3_bucket_link|

101 | 78 | users/78/"cv_Bob"

Why not keeping everithing int the MySQL databse(relational)?

since S3 Object Storage is especially for (images//PDFs/uploads)

- Store large files in S3 buckets

Why S3, not RDS/EBS?

- RDS is for structured relational data; ebs is like a disk tied to a machine while S3 is durable/cheap for objects

S3 -> for the React too

5.A) RDS (MySQL) (since relational entities, ACID)

- Use Amazon RDS MySQL for structured data: users/projects/orders/enrollments/job status.

Why RDS, not DB on EC2?

- rds handles backups, patching, monitoring, and failover options => low ops

6) ASYNC PROCESSING + NOTIFICATIONS

6.A) SQS Queue (background jobs for the !!ml tasks) with multiple workers

- push jobs to SQS (it will push only metadata, it will not push files!!!)
 - workers consume and process at controlled rate so backend does not crash
- Why SQS, not direct calls? to prevents overload/timeouts also buffers spikes

6.B) SNS Topic (notify multiple via email + webhook + logs)

- Publish "job finished" events to SNS

Why SNS?

- one event => many consumers cleanly.

6.C) SES Email (must email users since "notified via email when there is a change in the application process")

- Use SES to send emails: notification.

Why not app sending email => mail delivery/are hard and SES is built for it.

6.D) Step Functions (if the jobs for ml fails)

- Orchestrate pipeline: upload => review => process => publish with retries and error handling.

Why step functions -> a voids messy custom orchestration code inside backend.

7) NETWORKING + SECURITY

7.Core) VPC + Subnets + Security Groups

- we can deploy resources inside a VPC.
- public subnet: NAT if needed.
- private subnet ECS tasks and DB.
- security Groups DB SG: allow multiple inbounds

Why not IP allowlists? -> auto scaling changes IPs; SG-to-SG rules remain valid.

7.A) IAM Roles

- Assign IAM roles to compute (ECS) with least privilege:
 - S3 read/write
 - SQS send/receive
 - SNS publish
 - Secrets Manager read

why roles? since long-term credentials stored on servers.

7.B) Secrets Manager (we will have db creds/API keys/secrets exist)

- Store DB credentials and secrets in Secrets Manager; grant access via IAM role.

Why?

- Avoid hardcoding secrets in code/config.

8) OBSERVABILITY + SCALING + HA

8.A) Monitoring

- Use CloudWatch logs/metrics for errors, CPU/memory.

Why => troubleshooting and reliability.

8.B) Scaling

- ECS: scale task count based on CPU requests queue depth.

8.C) High Availability

- if we have relational DB enable Multi-AZ (RDS) when availability is needed

!!

I illustrated through the drawing some of the features, but due to the lack of time, not everything was added. I tried to explain as clear through bulletpoints and visually the HMRS app

!!

 [Screenshot 2026-01-15 at 15.49.08.png](#)

Comment:

Question **13**

Correct

Mark 0.50 out of 0.50

A media startup is developing a cloud-based content creation platform where users can design and share custom digital assets (such as templates, avatars, or widgets). The platform currently stores asset metadata in a relational database. The engineering team is considering migrating this data to Amazon DynamoDB. Which characteristics of the digital assets would make DynamoDB a better choice?

- ☐ Individual asset records commonly include very large binary files exceeding hundreds of megabytes
- ☐ Asset records follow a strict, predefined schema across all users
- ☒ Asset properties vary widely and differ from one user-created asset to another ✓
- ☐ Assets frequently require relational joins with multiple reference tables
- ☒ The system must scale automatically to support unpredictable creation and access patterns ✓

Your answer is correct.

Question **14**

Correct

Mark 0.50 out of 0.50

Documents are stored in S3 in a

Select one:

- ☐ a. Storage Account
- ☒ b. Bucket ✓
- ☐ c. Server

Your answer is correct.

Question **15**

Correct

Mark 0.50 out of 0.50

Simple Storage Service (S3) helps us

Select one:

- ☐ a. Create bootable virtual disks that can be attached to EC2 instances
- ☐ b. Host web applications
- ☒ c. Store and serve an infinite amount of objects such as images, documents, and videos 

Your answer is correct.

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